

RelaxedAsCoal

AML Challenge Report 2025/26

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1 Proposed Method

Our work began by studying the shared semantic space of text and images established by CLIP [1]. We leveraged this concept by using the CLIP latent space as an intermediate step in translating text embeddings into image embeddings. The developed model operates in two stages: an MLP first maps the initial roberta text embeddings (1024-dim) to the CLIP latent space (768-dim). This stage acts as a form of model distillation, guiding the MLP to retain only the text features relevant for image reconstruction. A second MLP then translates the resulting 768-dim embedding to the target DINOv2-Giant image embedding format (1536-dim). An additional subset of 30,000 samples from the COCO (Common Objects in Context) dataset [2] also was used. We ensured this subset had no overlap with the challenge test set to maintain data independence.

- **Architecture:**

Our Stage-1 MLP maps the 1024D input embedding to the 768D CLIP latent space. The architecture consists of three hidden layers with dimensions $1024 \rightarrow 1024 \rightarrow 896$. Each layer sequence follows the pattern: Linear $\rightarrow$ LayerNorm $\rightarrow$ GELU $\rightarrow$ Dropout. The output of the MLP is L2-normalized and combined with a learnable residual skip-connection, which is scaled by a trainable parameter (skip_weight).

Our Stage-2 MLP is used to transform the 768D CLIP embeddings into the 1536D DINOv2 image embedding space. It has the same architecture as Stage 1, the only difference being the dimensions of the hidden layers: $896 \rightarrow 1200 \rightarrow 1536$.

- **Loss Function:**

For Stage 1, we optimized the network using cosine similarity loss. This involved first L2-normalizing both the predicted embedding (from the Stage-1 MLP) and the target CLIP text embedding, and then computing the loss based on their similarity.

For Stage 2, we optimized the decoder using the combination of symmetric contrastive loss and a triplet loss. The symmetric contrastive loss computes bidirectional text-to-image and image-to-text cross-entropy over similarity logits scaled by a temperature parameter. The triplet loss uses hardest-

negative mining on normalized embeddings with a fixed margin. The final loss is a weighted sum of the two.

- **Training Details:**

The model was trained with AdamW for 30 epochs, using an initial warm-up followed by Cosine Annealing for learning rate management. Optimization was performed on minibatches using the loss function specified for each phase. Validation was carried out at every epoch, with the MRR@100 metric evaluated every two epochs. The model is saved when the validation loss is lower than the training loss, and an early stopping criterion with a patience of 5 epochs is applied.

2 Results and Discussion

Our two-stage architecture demonstrated superior performance due to its ability to leverage the CLIP latent space [1] as a structured intermediate representation.

The use of a non-linear adapter (Stage 1) to align the textual embeddings to the CLIP space significantly improves the translation by forcing the initial text features to conform to a robust structure known to accurately model the text-image relationship. This pre-aligned representation is then precisely mapped (Stage 2) into the final DINOv2 image embedding space by a second adapter.

This architectural strength, combined with training on a larger dataset [2], proved to be key to achieving better semantic alignment.

3 Conclusion

We developed a two-stage model that utilizes MLP adapters and contrastive/triplet loss to achieve superior alignment. By leveraging the intermediate CLIP latent space, the final model significantly outperformed the baseline, resulting in a higher MRR in the public evaluation and demonstrating the improved correspondence between captions and image embeddings. [Challenge Notebook](#)

What We Tried

Method 1: Latent Space Translation

The first method used was based on the ideas discussed in the paper [3]. We implemented a zero-shot latent translator by computing a closed-form affine mapping between two latent spaces. Given text and image embeddings, we solved a Procrustes problem using SVD to find an orthogonal rotation that best aligns them. With this model we were able to achieve an MRR of 0.78 on the public leaderboard.

Method 2: Deep 3-Layer MLP

We tested a deep MLP (3 hidden layers, GELU) hoping to capture complex non-linearities. This model performed significantly better than the first one, since it was capable of adding non-linear components to the calculations, increasing the complexity while increasing the performance. The new MRR achieved was of 0.84 in the public leaderboard.

Method 3: Flowing from Words to Pixels, CrossFlow

We also evaluated CrossFlow [4] by exploiting its continuous flow-matching formulation together with its dedicated text variational encoder, expecting the learned velocity fields to better align the encoded textual representations with the target latent space. Despite this, directly using the generated embeddings and projecting text embeddings to image embedding space yielded relatively low performance for this task, with an MRR of 0.75 on the test set.

Method 4: Additional training of intermediate translation to CLIP embeddings

This method further improved the performance of the MLP, by adding an intermediate translation for the text embeddings, therefore using 2 different MLPs. MLP 1 was trained to translate text embeddings to CLIP embeddings [1] and MLP 2 was trained to transform CLIP embeddings into image embeddings. This model achieved an MRR of 0.87.

Method 5: Increase of training dataset

The final improvement made was to increase the training dataset with the use of COCO [2]. With additional training the model was able to reach 0.90 in the public leaderboard MRR.

References

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